

**SHARP**

No.	LD-D5120Z02E
DATE	April 13 th , 2021

TECHNICAL LITERATURE

For

TFT-LCD module**These parts have corresponded with the RoHS directive.**MODEL No. **LQ140M1JW61**

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DEVELOPMENT DIVISION I
BUSINESS UNIT III
SHARP DISPLAY TECHNOLOGY CORPORATION

RECORDS OF REVISION

LQ140M1JW61

[illegible]



LD-D5120Z02E-1

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1. Application

This technical literature applies to a color TFT-LCD module, LQ140M1JW61.

2. Overview

ΔB

This module is a color active matrix LCD module incorporating **a-Si** TFT (Thin Film Transistor). **ΔB**

ΔD It is composed of a color TFT-LCD panel, driver ICs, a control circuit and power supply circuit, and a backlight unit. Graphics and texts can be displayed on a 1920×3×1080 dots panel with **16,777,216** colors by using eDP (Embedded Display Port) Ver. 1.2 to interface and supplying +3.3V DC supply voltage for TFT-LCD panel driving and applying +12V DC supply voltage for LED backlight driving.

In this TFT-LCD panel, color filters for excellent color performance and backlights for high brightness are incorporated to realize brighter and clearer pictures, making this model optimum for use in multi-media applications.

Backlight-driving LED controller is built in this module.

eDP Transfer rate Specification: 2.7Gbps / 2 lane

Color depth: 8 bit

3. Mechanical Specifications

ΔC

Parameter	Specifications	Unit
Display size	35.49 (13.97") Diagonal	cm
Active area	309.312(H)×173.988 (V)	mm
Pixel format	1920 (H)×1080 (V)	pixel
	(1 pixel = R+G+B dots)	
Pixel pitch	0.1611 (H) × 0.1611 (V)	mm
Pixel configuration	R,G,B vertical stripe	
Display mode	Normally black	
Surface treatment	Anti-glare, Hard coating (2H)	

Outline dimensions

Parameter		Min.	Typ.	Max.	Unit	Remark
Unit outline dimensions [Note 3-1]	Width	315.51	315.81	316.11	mm	w/o PWB
	Height	185.77	186.07	186.37	mm	
	Depth	-	2.80	3.0	mm	
Mass		-	-	285	g	

[Note 3-1] Outline dimensions is shown in Fig.2 (P.27)



4. Input Terminals

4 - 1. Symbol

CN1 (eDP signals and +3.3V DC power supply)

Pin No.	Symbol	I/O	Function	Remark
1	NC	-	No Connection	
2	H_GND	P	High Speed Ground	[Note 4-1-2]
3	Lane1_N	I	Complement Signal Link Lane 1	
4	Lane1_P	I	True Signal Link Lane 1	
5	H_GND	P	High Speed Ground	[Note 4-1-2]
6	Lane0_N	I	Complement Signal Link Lane 0	
7	Lane0_P	I	True Signal Link Lane 0	
8	H_GND	P	High Speed Ground	[Note 4-1-2]
9	AUX_CH_P	I	True Signal Auxiliary Channel	
10	AUX_CH_N	I	Complement Signal Auxiliary Channel	
11	H_GND	P	High Speed Ground	[Note 4-1-2]
12	VDD	P	LCD logic and driver power (3.3V)	
13	VDD	P	LCD logic and driver power (3.3V)	
14	NC	-	Reserved for LCD manufacturer's use	[Note 4-1-1]
15	LCD_GND	P	LCD logic and driver ground	
16	LCD_GND	P	LCD logic and driver ground	
17	HPD	O	HPD signal pin	[Note 4-1-3]
18	BL_GND	P	Backlight ground	
19	BL_GND	P	Backlight ground	
20	BL_GND	P	Backlight ground	
21	BL_GND	P	Backlight ground	
22	BL_ENABLE	I	Backlight On/Off	[Note 4-1-4]
23	BL_PWM_DIM	I	System PWM	[Note 4-1-5]
24	NC	-	Reserved for LCD manufacturer's use	[Note 4-1-1]
25	NC	-	Reserved for LCD manufacturer's use	[Note 4-1-1]
26	VBL	P	Backlight power	
27	VBL	P	Backlight power	
28	VBL	P	Backlight power	
29	VBL	P	Backlight power	
30	NC	-	Reserved for LCD manufacturer's use	[Note 4-1-1]

* P: Power, I: Input, O: Output

[Note 4-1-1] Don't input any signals or any powers into a NC pin. Keep the NC pin open.

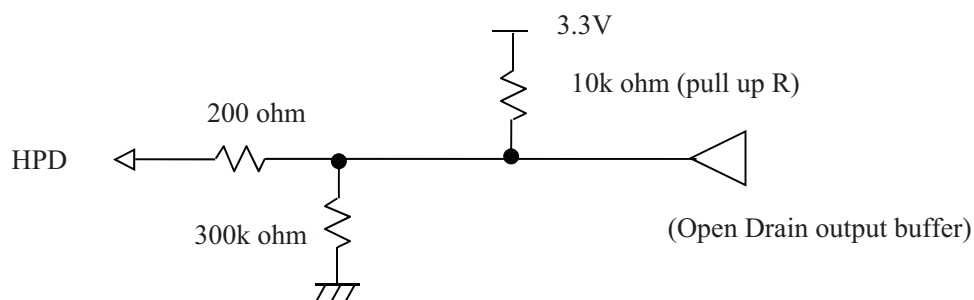
[Note 4-1-2] The shielding case is connected with signal GND.

•Connector used: MSAK24025P30 (STM)

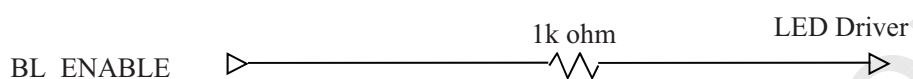
•Corresponding connector: 20453-030T or 20704-030T (I-PEX)

(Sharp is not responsible to its product quality, if the user applies a connector not corresponding to the above model.)

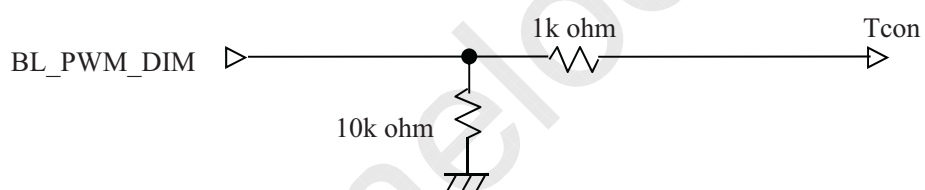
[Note 4-1-3] Output circuit is as below.



[Note 4-1-4] Input circuit is as below.



[Note 4-1-5] Input circuit is as below.



4 - 2. eDP interface

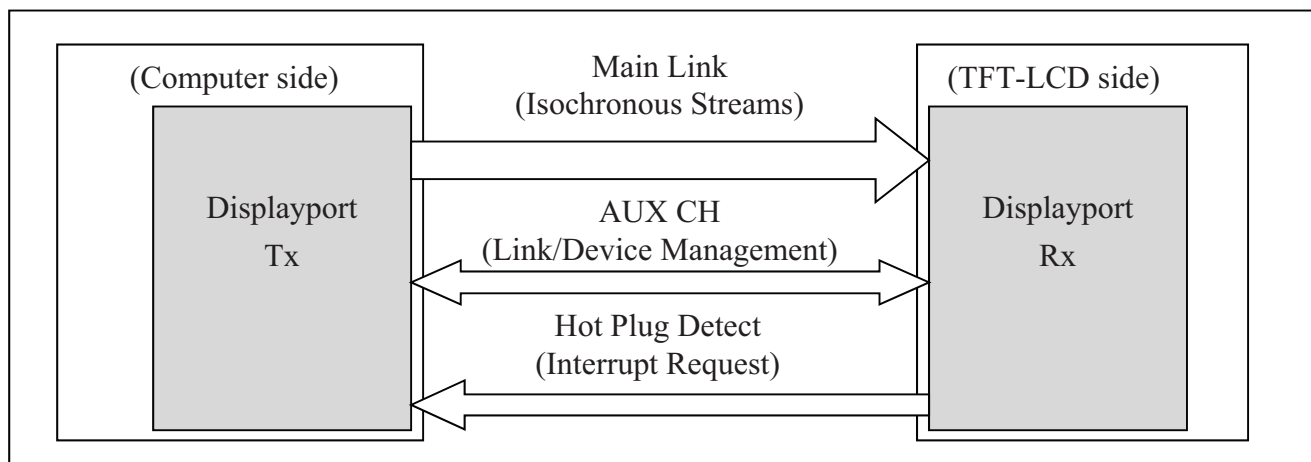


Fig.4-2-1 DP architecture.

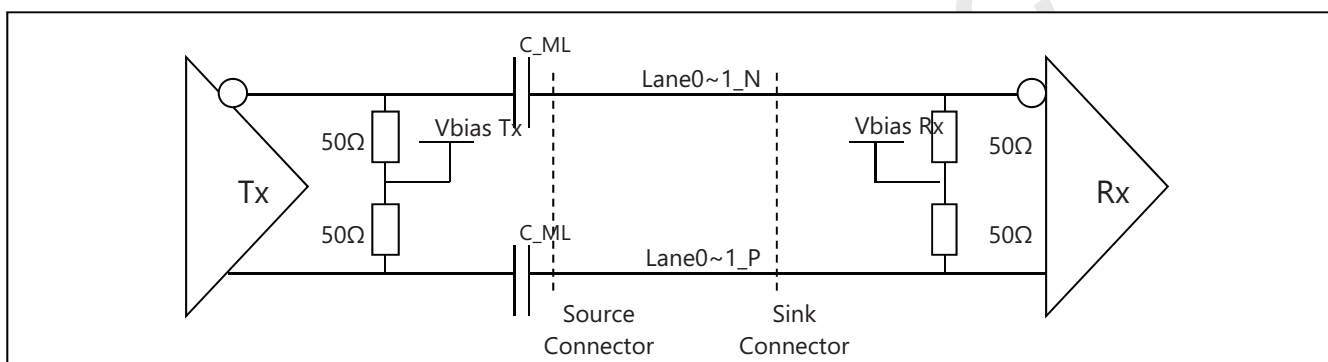


Fig.4-2-2 Main Link differential pair

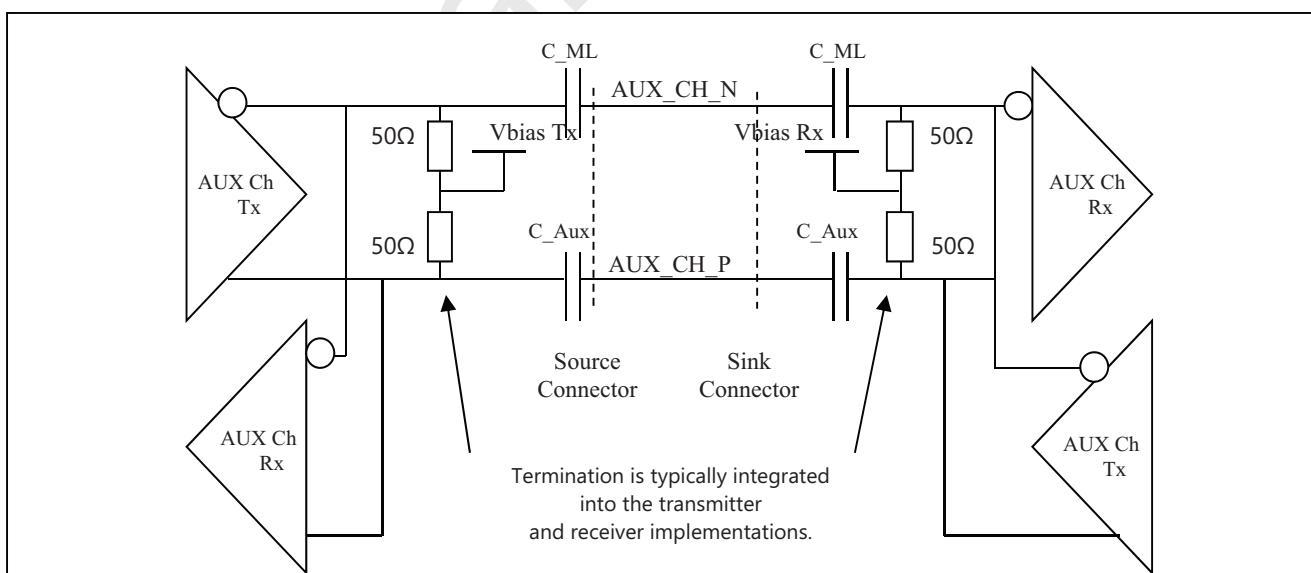


Fig.4-2-3 AUX Link differential pair



Table 4-2-1 eDP 2-lane 8-bit input data mapping.

 ΔD

Lane0	Lane1
R0-7:0	R1-7:0
G0-7:0	G1-7:0
B0-7:0	B1-7:0
R2-7:0	R3-7:0
G2-7:0	G3-7:0
B2-7:0	B3-7:0
R4-7:0	R5-7:0
G4-7:0	G5-7:0
B4-7:0	B5-7:0

5. Absolute Maximum Ratings

 ΔD

Parameter	Symbol	Condition	Ratings		Unit	Remark
			Min.	Max.		
+3.3V supply voltage	VDD	Ta=25°C	-0.3	+4.0	V	
Back Light supply voltage	VBL	Ta=25°C	-0.3	+26.5	V	
Input voltage(eDP)	VI_edp	Ta=25°C	-0.3	+1.5	V	[Note 5-1]
Input voltage (Logic)	VI_logic	Ta=25°C	-0.3	VDD+0.3	V	[Note 5-2]
Storage temperature (ambient)	Tstg	-	-25	+60	°C	[Note 5-3]
Operating temperature(ambient)	Topa	-	0	+50	°C	

[Note 5-1] eDP signals (Main Link, AUX)

[Note 5-2] BL_PWM_DIM and BL_EN

[Note 5-3] Humidity : 90%RH Max. at Ta≤+40°C

Maximum wet-bulb temperature at +39°C or less at Ta>+40°C.

No condensation.



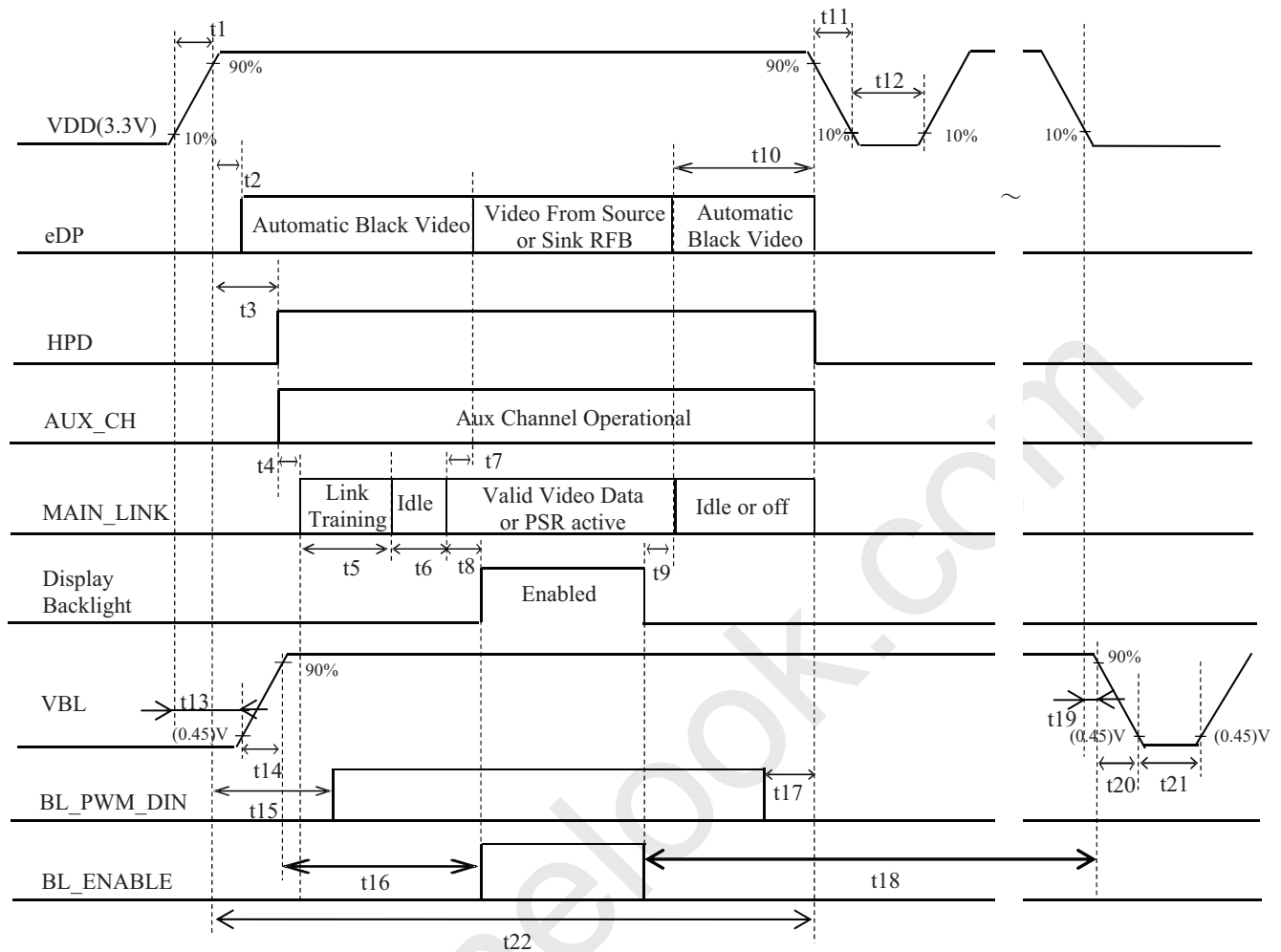
6. Electrical Characteristics

6 - 1. TFT-LCD panel driving

Ta=25°C

Parameter	Symbol	Min.	Typ.	Max.	Unit	Remark
DC Electrical Characteristics						
Supply voltage	VDD	+3.0	+3.3	+3.6	V	[Note 6-1-1]
Current dissipation (typ. pattern)	IDD ΔD	-	(207)	(259)	mA	[Note 6-1-11]
Logic power (Mosaic)	ΔD	-	(0.68)	(0.85)	W	[Note 6-1-11]
Permissive input ripple voltage	V _{RP}	-	-	100	mV _{P-P}	VDD=+3.3V
eDP HPD Signal Characteristics						
HPD High level output voltage	VOH _{HPD}	-	VDD-0.1	-	V	No Load
HPD Low level output voltage	VOL _{HPD}	-	0	-	V	No Load
eDP AUX Channel Characteristics						
Unit Interval for AUX channel	UIAUX	0.4	0.5	0.6	μs	
AUX peak-to-peak voltage at a transmitting device	VAUX-DIFF-pp	0.39	-	1.38	V	
AUX peak-to-peak voltage at a receiving device		0.32	-	1.36	V	
AUX DC common mode voltage	VAUX-DC-CM	0	-	2.0	V	
AUX short circuit current limit	IAUX_SHORT	-	-	90	mA	
AUX turn around common mode voltage	VAUX-TURN-CM	-	-	0.3	V	
AUX CH termination DC resistance	RAUX-_TERM	80	100	120	Ω	
Number of pre-charge pulses	Pre-charge Pulses	10	-	16	-	
eDP Main Link Receiver Characteristics						
Link clock down spreading	Down-Spread-Amplitude	0	-	0.5	%	
Differential Peak-to-peak input voltage at RX package pins	VRX-DIFFp-p	40	-	1200	mV	
RX DC Common Mode Voltage	VRX-DC-CM	0	-	2.0	V	
Differential termination resistance	VRX-TERM	80	100	120	Ω	
RX short circuit current limit	IRX-SHORT	-	-	50	mA	

[Note 6-1-1] ON-OFF conditions for supply voltage



[Note 6-1-2] Do not keep the interface signal high-impedance or unusual signal when power is on.

Symbol	Min	Max	Unit	Note
t1	0.5	10	ms	
t2	0	200	ms	
t3	0	200	ms	Sink AUX Channel must be optional upon HPD high.
t4	-	-	ms	
t5	-	-	ms	[Note 6-1-5]
t6	-	-	ms	[Note 6-1-5]
t7	0	50	ms	
t8	-	-	ms	[Note 6-1-6]
t9	-	-	ms	[Note 6-1-7]
t10	0	500	ms	
t11	1	50	ms	[Note 6-1-3]
t12	500	-	ms	[Note 6-1-4]
t13	-	-	ms	[Note 6-1-8]
t14	0.5	10	ms	
t15	100	-	ms	
t16	0	-	ms	[Note 6-1-9]
t17	0	-	ms	
t18	0	-	ms	[Note 6-1-9]
t19	-	-	ms	
t20	0.1	-	ms	
t21	100	-	ms	
t22	300	-	ms	

[Note 6-1-3] As for the power off sequence for VDD (t11), Be sure to keep above mentioned timing.

If the VDD power off sequence timing is other than shown above, LCD may cause permanent damage.

As for the power sequence for backlight, it is recommended to apply above mentioned input timing.

If the backlight is light on and off at a timing other than shown above, displaying image may get disturbed.

[Note 6-1-4] As for the power off-on sequence for VDD (t12), Be sure to keep above mentioned timing.

If the VDD power off-on sequence timing is other than shown above, LCD may cause permanent damage.

[Note 6-1-5] Dependent on the Source Link training protocol and Link idle.

[Note 6-1-6] The Source must assure display video is stable.

[Note 6-1-7] The Source must assure backlight is no longer illuminated.

[Note 6-1-8] Even if VBL Power is on earlier than VDD, It is not have any problem.

[Note 6-1-9] Prohibit the BL_ENABLE input of the period when it is not VBL Power on.

[Note 6-1-10] VDD-dip conditions

- 1) $2.5\text{ V} \leq \text{VDD} < 3.0\text{ V}$

$t_d \leq 10\text{ ms}$

Under above condition, the display image should return to an appropriate figure after VDD voltage recovers.

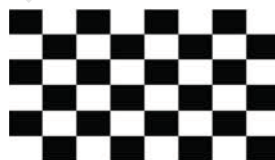
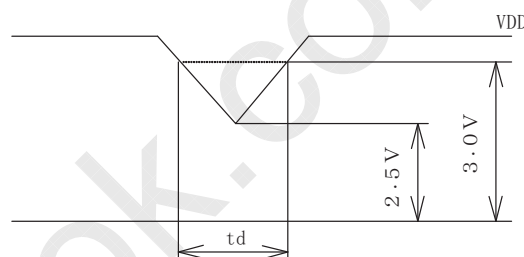
- 2) $\text{VDD} < 2.5\text{ V}$

VDD-dip conditions should also follow the ON-OFF conditions for supply voltage

[Note 6-1-11]

Typical current condition: mosaic pattern.

VDD=+3.3V



6 - 2. Backlight driving

The backlight system is an edge-lighting type with white-LED.

(It is usually required to measure under the following condition. : Ta=25°C±2°C)

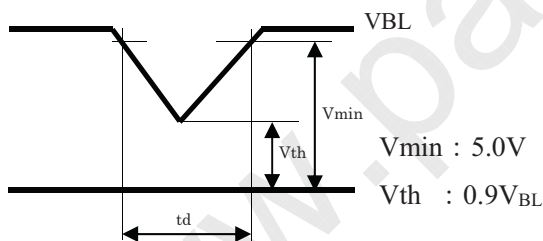
Parameter	Symbol	Min.	Typ.	Max	Unit	Remark
Supply voltage	VBL	5.0	12.0	21.0	V	
Current dissipation	I _{BL}	-	(166)	(175)	mA	VBL=12.0V
Power dissipation	P _{BL}	-	-	(2.10)	W	Duty Ratio=100%
Modulated light signal voltage	V _{PWMH}	1.85	-	3.6	V	BL_PWM_DIM
	V _{PWML}	0	-	0.7	V	
Brightness Control Duty Ratio	Duty	1	-	100	%	[Note6-2-1]
Brightness Control pulse width	T _{PWM}	5	-	-	μs	[Note6-2-2]
Brightness Control frequency	f _{PWM}	200	200	3300	Hz	
BL-EN signal voltage	V _{BL-ENH}	1.6	3.3	3.6	V	[Note6-2-3]
	V _{BL-ENL}	0	-	0.6	V	
Input signal pin current	I _{IN}	-	V _{IN} /400KΩ	-	μA	BL_EN BL_PWM_DIM
		-	V _{IN} /10KΩ	-		
LED lifetime	-	15,000	-	-	h	[Note6-2-4]

[Note6-2-1] PWM_Duty: 100%= Max luminance 1%= Min luminance

[Note6-2-2] The minimum value of the dimming signal pulse width is assumed regulations of the width of high and the width of low.

[Note6-2-3] V_{BL-EN} Input: High = BL turn on, Low or OPEN =BL turn off

VBL-dip conditions



1) $V_{th} \leq V_{BL} < V_{min}$: $t_d \leq 20 \text{ ms}$

2) $V_{BL} < V_{th}$: The condition of instantaneous voltage drop is applied to input voltage sequences

[Note6-2-4] Luminance becomes 50% of an initial value. (Ta=25°C, PWM=100%)

7. Timing Characteristics of Input Signals

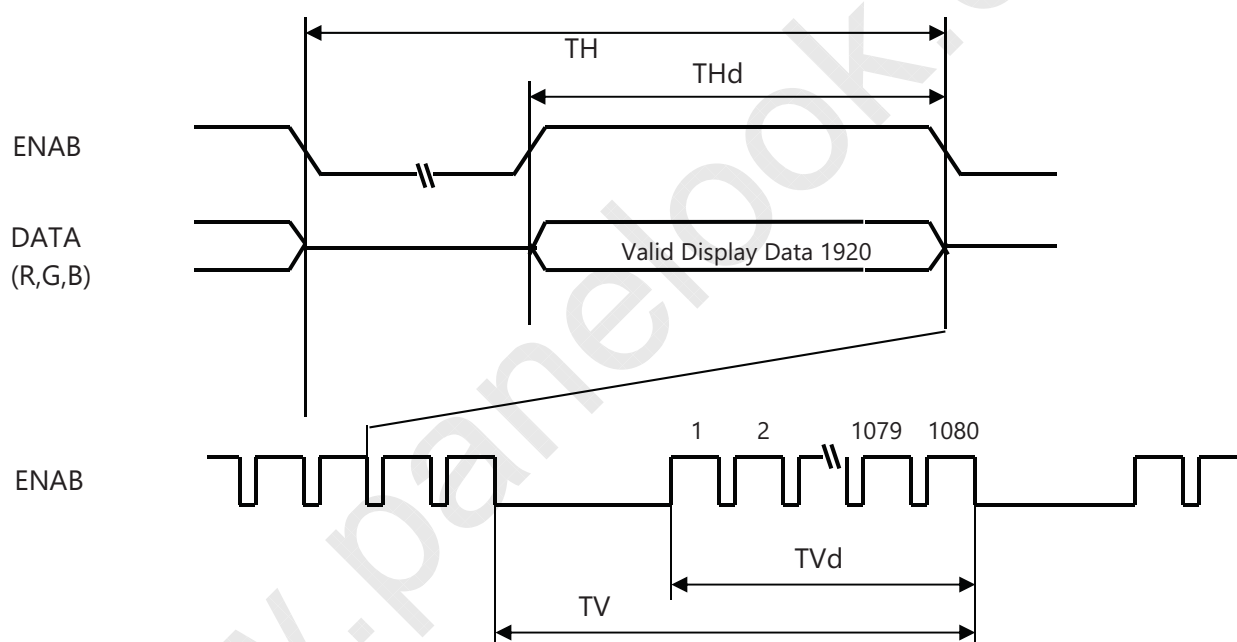
△D △E

7-1. Timing characteristics

VDD=+3.0V~+3.6V, Ta=0°C~+50°C

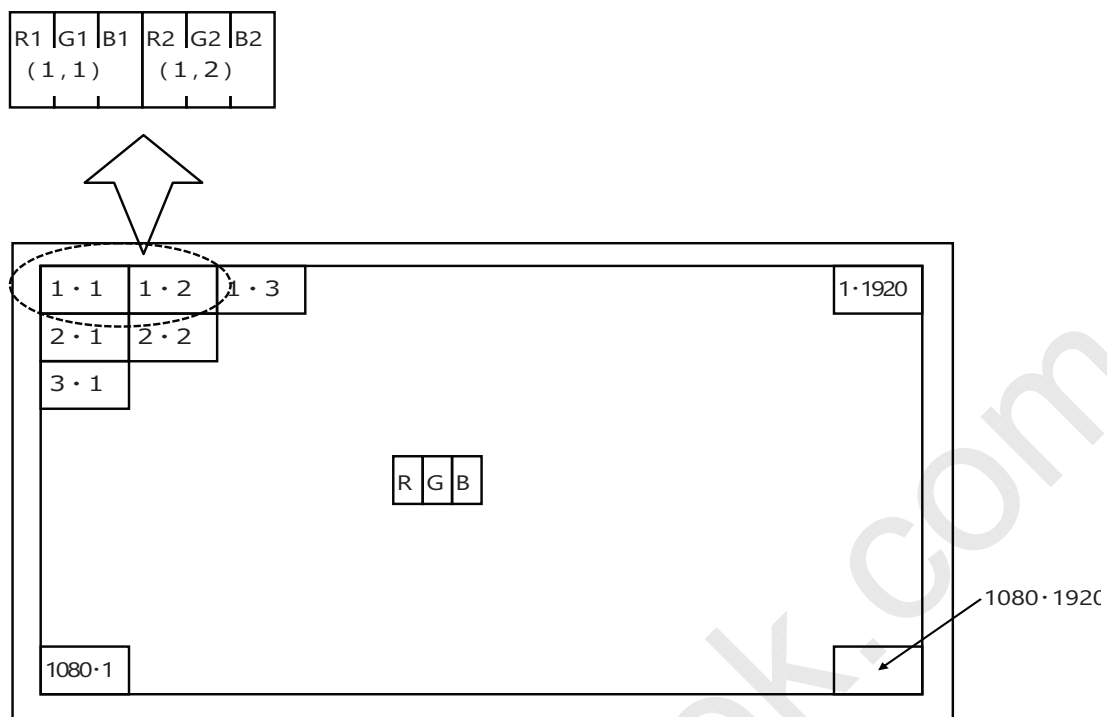
	Parameter	Symbol	Min.	Typ.	Max.	Unit	Remark
Clock	Frequency	1/Tc	136.53	138.53	140.53	MHz	[Note 7-1-1]
Data enable Signal	Horizontal period	TH	2050	2080	2210	clock	
			-	15.015	-	μs	
	Horizontal period (High)	THd	-	1920	-	clock	
	Vertical period	TV	1096	1110	1124	line	
			-	16.667	-	ms	
	Vertical period (High)	TVd	-	1080	-	line	

[Note 7-1-1] In case of using the long vertical period, the deterioration of display quality, flicker, etc, may occur.



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7 - 2. Input data signals and display position on the screen



Display position of input data (V·H)



8. Input Signals, Basic Display Colors and Gray Scale of Each Color

△D

	Colors & Gray Scale	Date signal																											
		Gray Scale	R0	R1	R2	R3	R4	R5	R6	R7	G0	G1	G2	G3	G4	G5	G6	G7	B0	B1	B2	B3	B4	B5	B6	B7			
			LSB							MSB							LSB							MSB					
Basic Color	Black	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Blue	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1		
	Green	-	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0		
	Cyan	-	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
	Red	-	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Magenta	-	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1		
	Yellow	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0		
	White	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
Gray Scale of Red	Black	GS0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↑	GS1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Darker	GS2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↑	↓	↓							↓							↓												
	↓	↓	↓							↓							↓												
	Brighter	GS253	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↓	GS254	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Red	GS255	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Gray Scale of Green	Black	GS0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↑	GS1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	Darker	GS2	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↑	↓	↓							↓							↓												
	↓	↓	↓							↓							↓												
	Brighter	GS253	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0		
	↓	GS254	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0		
	Green	GS255	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0		
Gray Scale of Blue	Black	GS0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	↑	GS1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0		
	Darker	GS2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0		
	↑	↓	↓							↓							↓												
	↓	↓	↓							↓							↓												
	Brighter	GS253	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	1		
	↓	GS254	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1		
	Blue	GS255	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1		

0 : Low level voltage, 1 : High level voltage

Each basic color can be displayed in 256 gray scales from 8 bit data signals.

According to the combination of 24 bit data signals, the 16,777,216-colors display can be achieved on the screen.



9. EDID Specifications

9 - 1. EDID data structure

△D △E

This is the EDID(Extended Display Identification Data) data formats to support displays as defined in the VESA Plug & Display

Byte (decimal)	Byte (hex)	Field Name and Comments	Value (hex)	Value (binary)
0	00	Header	00	00000000
1	01	Header	FF	11111111
2	02	Header	FF	11111111
3	03	Header	FF	11111111
4	04	Header	FF	11111111
5	05	Header	FF	11111111
6	06	Header	FF	11111111
7	07	Header	00	00000000
8	08	EISA manufacture code = SHP	4D	01001101
9	09	EISA manufacture code (Compressed ASCII)	10	00010000
10	0A	Product code (LQ140M1JW61 : 5440)	40	01000000
11	0B	Product code (hex, LSB first)	15	00010101
12	0C	LCD module Serial No (fixed "0")	00	00000000
13	0D	LCD module Serial No (fixed "0")	00	00000000
14	0E	LCD module Serial No (fixed "0")	00	00000000
15	0F	LCD module Serial No (fixed "0")	00	00000000
16	10	Week of manufacture	0A	00001010
17	11	Year of manufacture - 1990 (ex 2000 - 1990 = 10)	1F	00011111
18	12	EDID structure version # = 1	01	00000001
19	13	EDID revision # = 4	04	00000100
20	14	Video i/p definition = Digital 6bit DP support	A5	10100101
21	15	Max H image size(cm) = 31cm	1F	00011111
22	16	Max V image size(cm) = 17cm	11	00010001
23	17	Display gamma (2.2×100) - 100 = 120	78	01111000
24	18	Feature support(standby, suspend, RGB color/Prefer Time)	02	00000010
25	19	Red/Green Low bit(RxRy/GxGy)	3A	00111010
26	1A	Blue/White Low bit(BxBy/WxWy)	A0	10100000
27	1B	Red X(Rx) (written value 0.59)	97	10010111
28	1C	Red Y(Ry) (written value 0.355)	5A	01011010
29	1D	Green X(Gx) (written value 0.346)	58	01011000
30	1E	Green Y(Gy) (written value 0.545)	8B	10001011
31	1F	Blue X(Bx) (written value 0.155)	27	00100111
32	20	Blue Y(By) (written value 0.1)	19	00011001
33	21	White X(Wx) (written value 0.313)	50	01010000
34	22	White Y(Wy) (written value 0.329)	54	01010100
35	23	Established timings 1	00	00000000
36	24	Established timings 2	00	00000000
37	25	Established timings 3(Manufacture's reserved timing)	00	00000000
38	26	Standard timing ID1	01	00000001
39	27	Standard timing ID1	01	00000001
40	28	Standard timing ID2	01	00000001
41	29	Standard timing ID2	01	00000001
42	2A	Standard timing ID3	01	00000001
43	2B	Standard timing ID3	01	00000001
44	2C	Standard timing ID4	01	00000001
45	2D	Standard timing ID4	01	00000001
46	2E	Standard timing ID5	01	00000001
47	2F	Standard timing ID5	01	00000001
48	30	Standard timing ID6	01	00000001
49	31	Standard timing ID6	01	00000001
50	32	Standard timing ID7	01	00000001
51	33	Standard timing ID7	01	00000001
52	34	Standard timing ID8	01	00000001
53	35	Standard timing ID8	01	00000001
54	36	Detailed timing descriptor#1 fck/10000 (=138.53MHz/10000=13853=361Dh)	1D	00011101
55	37	#1 fck	36	00110110
56	38	#1 Horizontal active 1920=780h 80h	80	10000000
57	39	#1 Horizontal blanking 160=0A0h A0h	A0	10100000
58	3A	#1 Horizontal active/Horizontal blanking 70h	70	01110000
59	3B	#1 Vertical active 1080=438h 38h	38	00111000
60	3C	#1 Vertical blanking 30=01Eh 1Eh	1E	00011110



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61	3D	#1 Vertical active/Vertical blanking 40h	40	01000000
62	3E	#1 Horizontal sync, offset(Thfp) 48=030h 30h	30	00110000
63	3F	#1 Horizontal sync, width 32=020h 20h	20	00100000
64	40	#1 Vertical sync, offset/Vertical sync, width (offset=3h/width=5h)	35	00110101
65	41	#1 Horizontal sync offset/width/Vertical sync offset/width	00	00000000
66	42	#1 Horizontal image size 309.312mm=135h 35h	35	00110101
67	43	#1 Vertical image size 173.988mm=0AEh AEh	AE	10101110
68	44	#1 Horizontal image size/Vertical image size 10h	10	00010000
69	45	Horizontal border	00	00000000
70	46	Vertical border	00	00000000
71	47	Flags(Non-interlaced=0/non 3D=00/Digital separate=11/Horizontal polarity/Vertical	18	00011000
72	48	Detailed timing descriptor#2	00	00000000
73	49	Flag	00	00000000
74	4A	Reserved	00	00000000
75	4B	Dummy Descriptor	10	00010000
76	4C	Flag	00	00000000
77	4D	1st Dummy	00	00000000
78	4E	2nd Dummy	00	00000000
79	4F	3rd Dummy	00	00000000
80	50	4th Dummy	00	00000000
81	51	5th Dummy	00	00000000
82	52	6th Dummy	00	00000000
83	53	7th Dummy	00	00000000
84	54	8th Dummy	00	00000000
85	55	9th Dummy	00	00000000
86	56	10th Dummy	00	00000000
87	57	11th Dummy	00	00000000
88	58	New line character #2 indicates end	00	00000000
89	59	Padding with 'blank' character	00	00000000
90	5A	Detailed timing descriptor#3	00	00000000
91	5B	Flag	00	00000000
92	5C	Reserved	00	00000000
93	5D	Dummy Descriptor	10	00010000
94	5E	Flag	00	00000000
95	5F	1st Dummy	00	00000000
96	60	2nd Dummy	00	00000000
97	61	3rd Dummy	00	00000000
98	62	4th Dummy	00	00000000
99	63	5th Dummy	00	00000000
100	64	6th Dummy	00	00000000
101	65	7th Dummy	00	00000000
102	66	8th Dummy	00	00000000
103	67	9th Dummy	00	00000000
104	68	10th Dummy	00	00000000
105	69	11th Dummy	00	00000000
106	6A	New line character #3 indicates end	00	00000000
107	6B	Padding with 'blank' character	00	00000000
108	6C	Detailed timing descriptor #4	00	00000000
109	6D	Flag	00	00000000
110	6E	Reserved	00	00000000
111	6F	Display Product name	FE	11111110
112	70	Flag	00	00000000
113	71	Supplier P/N#1 (L)	4C	01001100
114	72	Supplier P/N#2 (Q)	51	01010001
115	73	Supplier P/N#3 (1)	31	00110001
116	74	Supplier P/N#4 (4)	34	00110100
117	75	Supplier P/N#5 (0)	30	00110000
118	76	Supplier P/N#6 (M)	4D	01001101
119	77	Supplier P/N#7 (1)	31	00110001
120	78	Supplier P/N#8 (J)	4A	01001010
121	79	Supplier P/N#9 (W)	57	01010111
122	7A	Supplier P/N#10 (6)	36	00110110
123	7B	Supplier P/N#11 (1)	31	00110001
124	7C	Supplier P/N#12 ("Space")	0A	00001010
125	7D	(If<13 char, then terminate with ASCII code 0Ah, set remaining char 20h)	20	00100000
126	7E	Extension flag	00	00000000
127	7F	Checksum	2C	00101100

10. Optical Characteristics

Ta=+25°C, VDD=3.3V, VBL = +12.0V

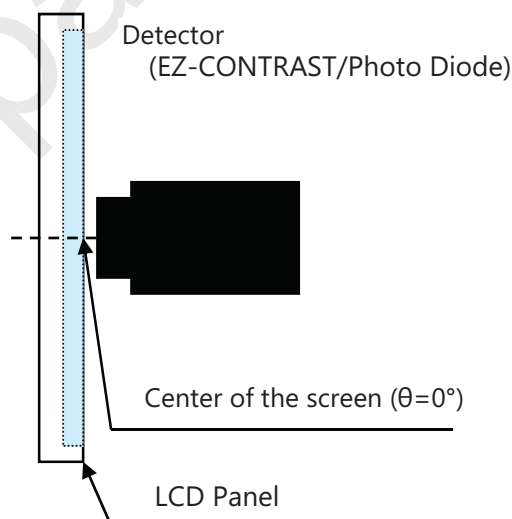
Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing angle range	Horizontal	θ21, θ22	CR>10	80	85	-	deg.	[Note 10-1, 10-3, 10-4, 10-6]
	Vertical	θ11, θ12		80	85	-	deg.	
Contrast ratio		CR	θ=0°	700	1000	-		[Note 10-2, 10-4, 10-6]
Response time		τr+τd	θ=0°	-	25	35	ms	[Note 10-1, 10-5, 10-6]
Chromaticity of white	x	x		Typ. -0.03	0.313	Typ. +0.03		[Note 10-2, 10-6] Normal operation (PWM Duty=100%)
	y	y			0.329			
Chromaticity of red	X	X			(0.590)			
	y	y			(0.355)			
Chromaticity of green	x	x			(0.346)			
	y	y			(0.545)			
Chromaticity of blue	x	x			(0.155)			
	y	y			(0.100)			
NTSC ratio		%		-	45	-		[Note 10-2, 10-8] Normal operation (PWM Duty=100%)
Luminance of white		Y _{LI}		212	250	-	cd/m ²	
White uniformity		δW	θ=0°	-	-	1.65		[Note 10-2, 10-7]

※ The measurement shall be taken 30 minutes after lighting the module at the following rating:

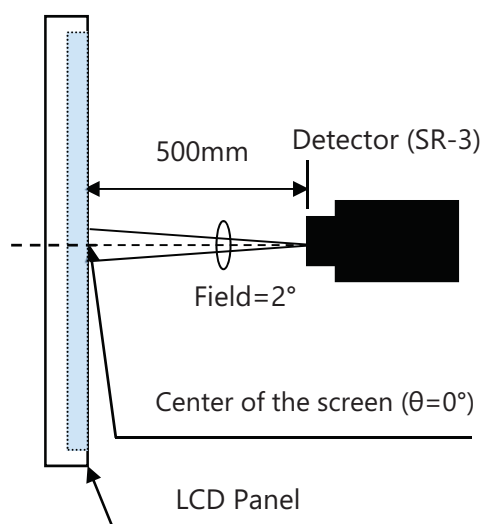
Condition: PWM Duty = 100%

The optical characteristics shall be measured in a dark room or equivalent.

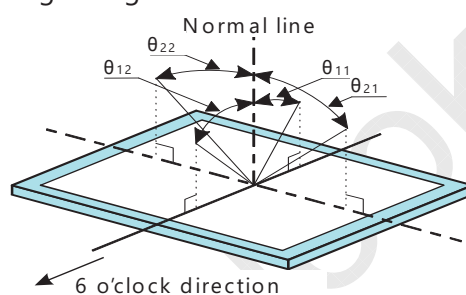
[Note 10-1] Measurement of viewing angle range and Response time.



[Note 10-2] Measurement of luminance and Chromaticity and Contrast.



[Note 10-3] Definitions of viewing angle range:



[Note 10-4] Definition of contrast ratio:

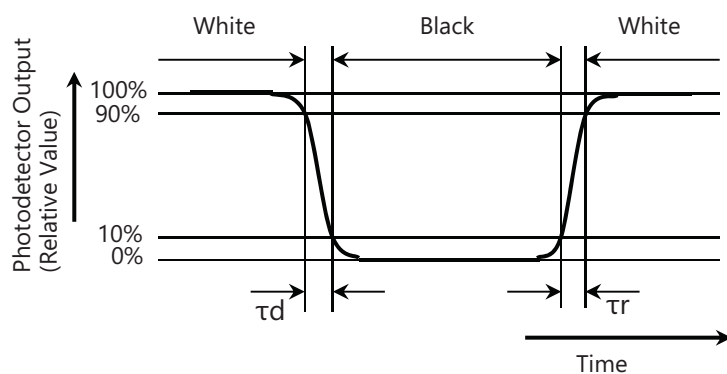
The contrast ratio is defined as the following.

$$CR = \frac{\text{Luminance (brightness) all pixels White}}{\text{Luminance (brightness) all pixels Black}}$$

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[Note 10-5] Definition of response time:

The response time is defined as the following figure and shall be measured by switching the input signal for "black" and "white".

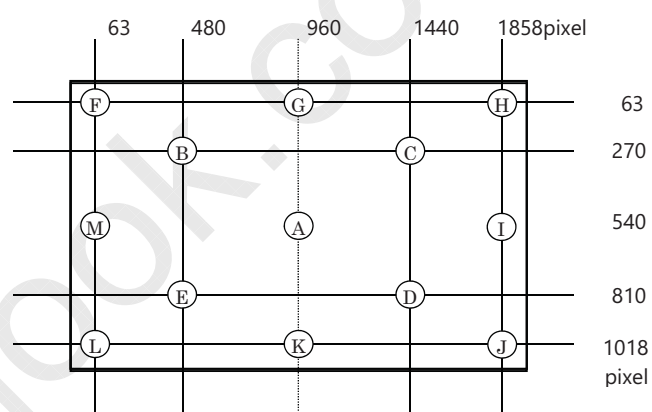


[Note 10-6] This shall be measured at center of screen.

[Note 10-7] Definition of white uniformity:

White uniformity is defined as the

following with thirteen measurements (A~M).



$$\delta_{w13} = \frac{\text{Maximum Luminance of thirteen points (brightness)}}{\text{Minimum Luminance of thirteen points (brightness)}}$$

[Note 10-8] Luminance of white is defined as the average of the above five points brightness (A~E).

11. Display Quality

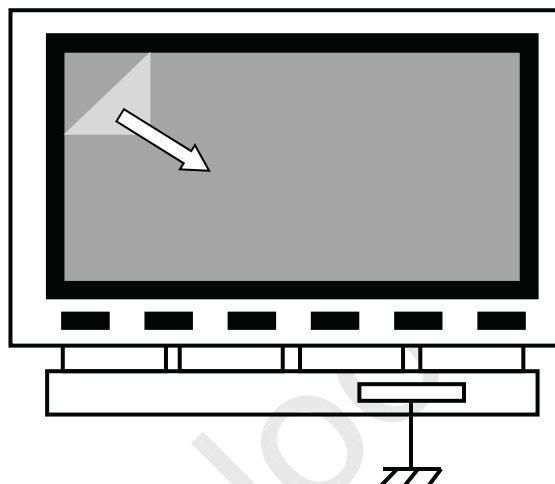
The display quality of the color TFT-LCD module shall be in compliance with the Incoming Inspection Standard.

12. Handling Precautions

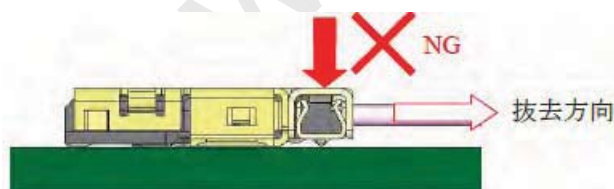
- a) Be sure to turn off the power supply when inserting or disconnecting the cable.
Please insert for too much stress not to join a connector in the case of insertion of a connector.
- b) Be sure to design the cabinet so that the module can be installed without any extra stress such as warp or twist.
- c) Since the front polarizer is easily damaged, pay attention not to scratch it.
- d) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- e) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
- f) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
Handle with care.
- g) Since CMOS LSI is used in this module, take care of static electricity and injure the human earth when handling. Observe all other precautionary requirements in handling components.
- h) This module has its circuitry PCBs on the side and should be handled carefully in order not to be stressed.
- i) Protect sheet (Laminate film) is attached to the module surface to prevent it from being scratched. Peel the sheet off slowly just before the use with strict attention to electrostatic charges. Ionized air shall be blown over during the action. Blow off the 'dust' on the polarizer by using an ionized nitrogen gun, etc. Working under the following environments is desirable.
 - All workers wear conductive shoes, conductive clothes, conductive fingerstalls and grounding belts without fail.
 - Use Ionized blower for electrostatic removal, and peel of the protect sheet with a constant speed. (Peeling of it at over 2 seconds)
- j) The polarizer surface on the panel is treated with anti-glare. In case of attaching protective board over the LCD, be careful about the optical interface fringe etc. which degrades display quality.
- k) Do not expose the LCD module to a direct sunlight, for a long period of time to protect the module from the ultraviolet ray.
- l) When handling LCD modules and assembling them into cabinets, please be noted that long-term storage in the environment of oxidization or deoxidization gas and the use of such materials as reagent, solvent, adhesive, resin, etc. which generate these gasses, may cause corrosion and discoloration of the LCD modules.
- m) Liquid crystal contained in the panel may leak if the LCD is broken. Rinse it as soon as possible if it gets inside your eye or mouth by mistake.
- n) Disassembling the module can cause permanent damage and should be strictly avoided.
Please don't remove the fixed tape, insulating tape etc. that was pasted on the original module. (Except for protection film of the panel.)
- o) Be careful when using it for long time with fixed pattern display as it may cause afterimage. (Please use a screen saver etc., in order to avoid an afterimage.)
- p) If a minute particle enters in the module and adheres to an optical material, it may cause display non-uniformity issue, etc. Therefore, fine-pitch filters must be installed to cooling and inhalation hole if you intend to install a fan.
- q) Epoxy resin (amine series curing agent), silicone adhesive material (dealcoholization series and oxime series), tray forming agent (azo compound) etc., in the cabinet or the packing materials may induce abnormal display with polarizer film deterioration regardless of contact or noncontact to polarizer film.
Be sure to confirm the component of them.
- r) Do not use polychloroprene. If you use it, there is some possibility of generating Cl₂ gas that influences the reliability of the connection between LCD panel and driver IC.

- s) Do not put a laminate film on LCD module, after peeling of the original one. If you put on it, it may cause discoloration or spots because of the occurrence of air gaps between the polarizer and the film.
- t) Ground module bezel to stabilize against EMI and external noise.
- u) When you peel off the protection film,
- Be sure to peel off slowly (recommended more than 5sec) and constant speed.
 - Peeling direction shows the following Figure.
 - Be sure to ground person with adequate methods such as the anti-static wrist band.
 - Be sure to ground all terminals of the PWB connector while peeling of the protection film.
 - Ionized air should be blown over during peeling action.
 - The protection film must not touch driver-ICs, PWB and all components on PWB.
 - If adhesive may remain on the polarizer after the protection film peeling off, please remove with isopropyl-alcohol.

Front view



- v) The glass of display is bigger than Back-light, so please be careful of below points (especially about 4 corners.)
- Don't hit LCD module with hard solid to prevent glass cracking.
 - Please wear the gloves etc. to protect hands when handling LCD module.
- w) If you apply downward force from connector above especially edge of pull bar when you un-plug from receptacle, there is a possibility to damage PCB like picture below. This downward force on edge of pull-bar will be caused of short and circuit and disconnection of PCB layout. Therefore, please don't apply and load from above.



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12 - 1. Precautions for Handling Tray

- a.) Hold center of short side of tray with both hands when handling one or more trays.



Caution : Do not handle with only one hand .



- b.) Always place tray on flat surface.
- c.) When stacking trays, please align same shape corner of each tray.
One corner is R corner. (Ref. Pic. 1)
- d.) Maximum stacking quantity is the number of trays inside one box.

Ex.: In case of 30pcs LCD module per box (2pcs LCD module per tray), maximum stacking is 15 trays.



R Corner

Pic.1 Stacking tray

13. Packaging Condition

Piling number of cartons	8
Package quantity in one carton	20
Carton size	544 x 438 x 161 mm
Total mass of one carton filled with full modules	11 kg
Packing form	Fig.1

14. Storage Conditions

a) Environmental condition range of storage temperature and humidity.

Temperature: 0~40 °C

Relative humidity: 0~95%

[Note] Please refer below as a mean value of the environmental conditions.

Summertime: temperature 20 to 35 degrees Celsius, humidity, 85% and below

Wintertime: temperature 5 to 15 degrees Celsius, humidity, 85% and below

Please maintain within 240 hours of accumulated length of storage time, with conditions of 40 degrees Celsius and room humidity of 95%.

b) Direct sun light

Please keep the product in a dark room or cover the product to protect from direct sun light.

c) Atmospheric condition

Please refrain from keeping the product with possible corrosive gas or volatile flux.

d) Prevention of dew

Please store the product carton either on a wooden pallet or a stand / rack to prevent dew.

Do not place directly on the floor. In addition, to obtain moderate ventilation in between the pallet's top and bottom surfaces, pile the cartons up in a single direction and in order.

Please place the product cartons away from the storage wall.

e) Storage period

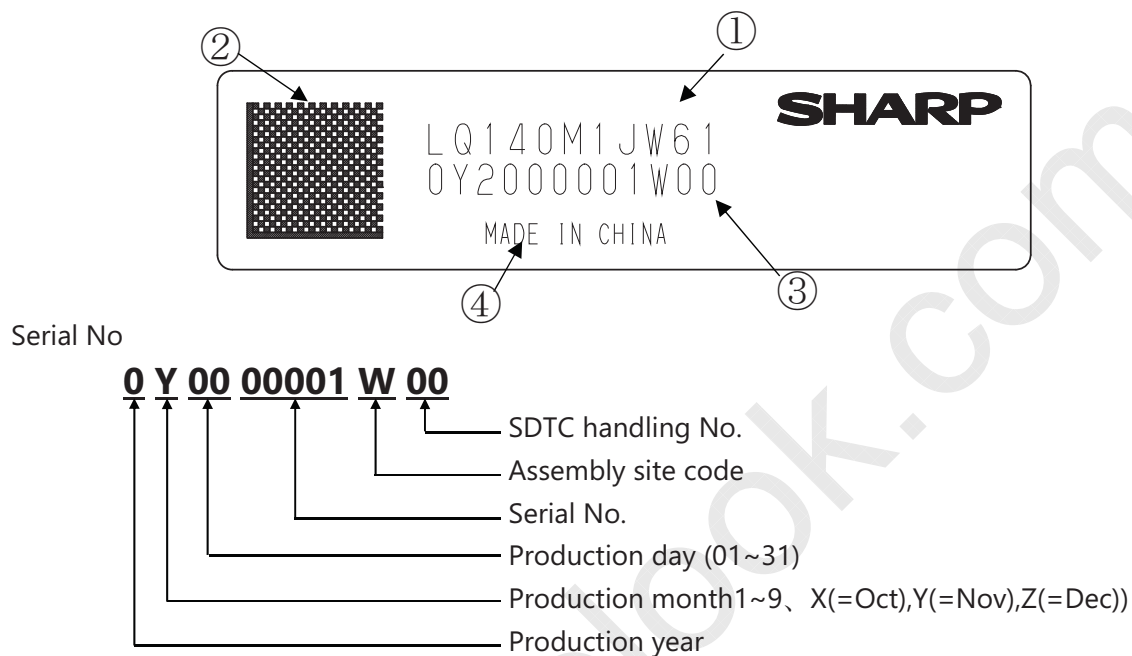
Within above mentioned conditions, maximum storage period should be one year.

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15. Label

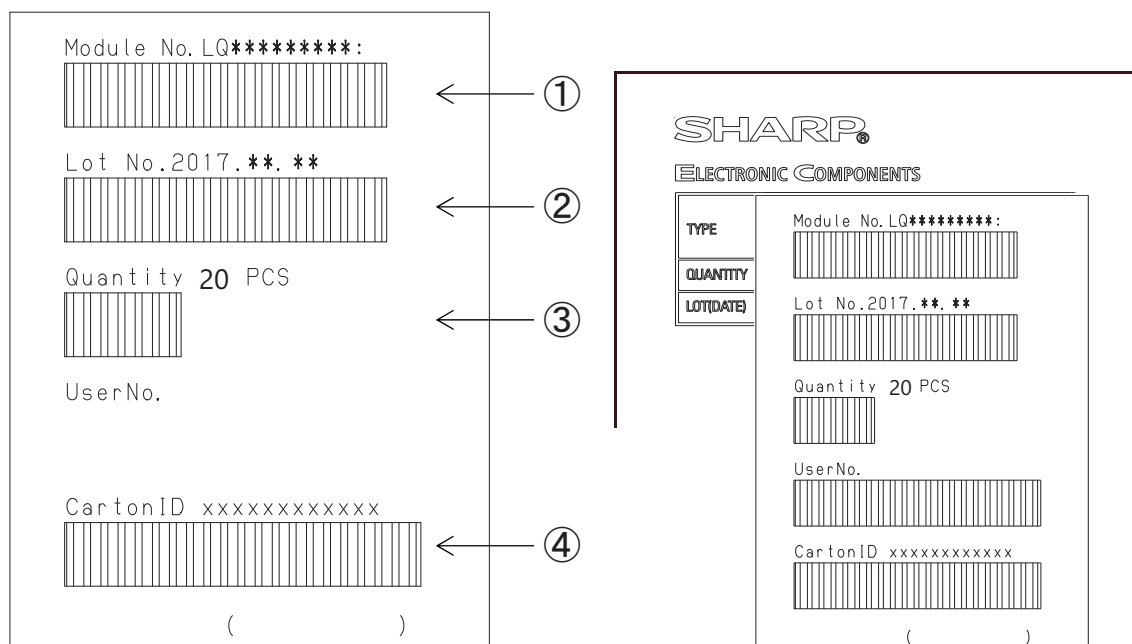
1) Module barcode label:

- ① Model No.
- ② 2D Barcode (①Model No. + ③Serial number)
- ③ Serial No.
- ④ Country of manufacture



2) Packing bar code label

- ① Model No.
- ② date
- ③ Quantity (20pcs / Carton)
- ④ Carton ID





16. RoHS Directive

This LCD module is compliant with RoHS Directive.

17. Reliability Test Items

No.	Test item	Conditions
1	High temperature storage test	Ta = 60°C 240h
2	Low temperature storage test	Ta = -20°C 240h
3	High temperature & high humidity operation test	Ta = 40°C; 90%RH 240h (No condensation)
4	High temperature operation test	Ta = 50°C 240h
5	Low temperature operation test	Ta = 0°C 240h
6	Vibration test (non- operating)	Frequency: 10~57Hz/Vibration width (one side):0.076mm 57~500Hz/Acceleration:9.8m/s ² Sweep time: 11minutes Test period: 1 hour for each direction of X, Y, Z
7	Shock test (non- operating)	Max. gravity: 490 m/s ² Pulse width: 11 ms, half sine wave Direction : ±X,±Y,±Z once for each direction.
8	ESD	Panel center, Around the module : One time for each position Contact discharging method: C=150pF, R=330Ω (non-operating) Pass ± 10kV (operating) Pass ± 8kV Air discharge method: C=150pF, R=330Ω (non-operating) Pass ± 20kV (operating) Pass ± 15kV If LCM can self-recover, temporary degradation of performance is allowed.

[Result Evaluation Criteria] Under the display quality test conditions with normal operation state. Do not change these conditions as such changes may affect practical display function.

[Normal operation state] Temperature: +15~+35°C, Humidity: 45~75%, Atmospheric pressure: 86~106kPa

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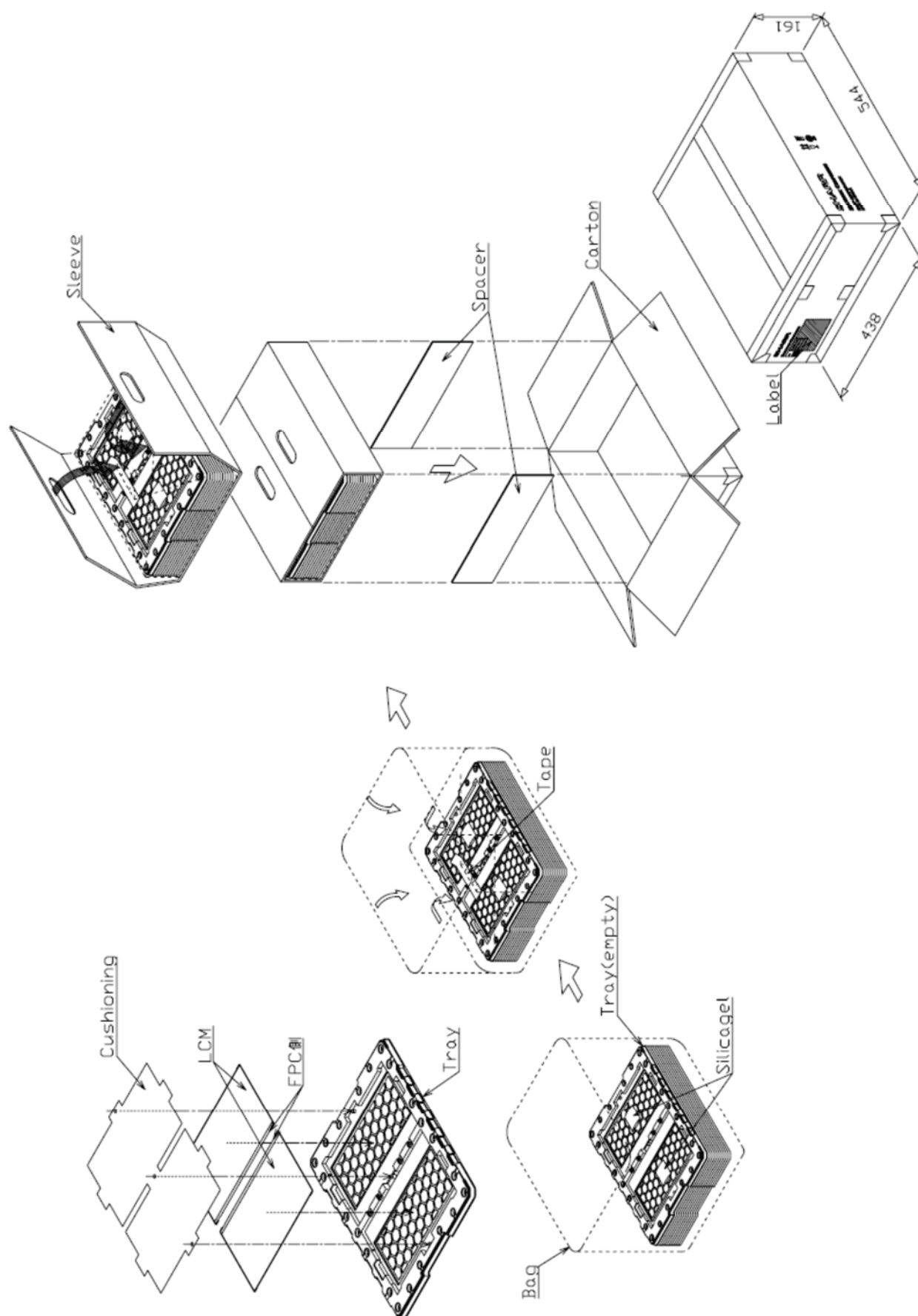


Fig. 1 Packaging Condition



Fig. 2 Outline Dimensions